

Null Mutations of the *Dictyostelium* Cyclic Nucleotide Phosphodiesterase Gene Block Chemotactic Cell Movement in Developing Aggregates

Richard Sucgang,^{*,1} Cornelis J. Weijer,[†] Florian Siegert,[‡] Jakob Franke,^{*} and Richard H. Kessin^{*}

^{*}Department of Anatomy and Cell Biology, College of Physicians and Surgeons, Columbia University, 630 West 168 Street, New York, New York 10032; [†]Department of Anatomy and Physiology, Old Medical School, University of Dundee, Dundee DD1 4HN, Scotland; and [‡]Zoologisches Institut, Universität München, Luisenstrasse 14, 80333 Munich, Germany

Extracellular cAMP is a critical messenger in the multicellular development of the cellular slime mold *Dictyostelium discoideum*. The levels of cAMP are controlled by a cyclic nucleotide phosphodiesterase (PDE) that is secreted by the cells. The PDE gene (*pdsA*) is controlled by three promoters that permit expression during vegetative growth, during aggregation, and in prestalk cells of the older structures. Targeted disruption of the gene aborts development, and complementation with a modified *pdsA* restores development. Two distinct promoters must be used for full complementation, and an inhibitory domain of the PDE must be removed. We took advantage of newly isolated PDE-null cells and the natural chimerism of the organism to ask whether the absence of PDE affected individual cell behavior. PDE-null cells aggregated with isogenic wild-type cells in chimeric mixtures, but could not move in a coordinated manner in mounds. The wild-type cells move inward toward the center of the mound, leaving many of the PDE-null cells at the periphery of the aggregate. During the later stages of development, PDE-null cells in the chimera segregate to regions which correspond to the prestalk region and the rear of the slug. Participation in the prespore/spore population returns with the restoration of a modified *pdsA* to the null cells. © 1997 Academic Press

INTRODUCTION

During *Dictyostelium discoideum*'s development, starving cells generate pulses of extracellular cAMP which create temporal gradients that attract other cells. *Dictyostelium* amoebae respond to the pulses of cAMP in two ways: by migrating to higher concentrations of cAMP and by secreting more cAMP. This results in propagating waves of cAMP that guide the cells during aggregation to form mounds of approximately 10⁵ cells; within these mounds, the cells undergo a series of movements to establish a tipped mound, a slug, and finally, a mature fruiting body. Waves of cAMP also occur in the multicellular structures (Steinbock *et al.*, 1993) and can be observed as propagating

optical density waves brought about by periodic changes in cell shape in response to cAMP (Siegert and Weijer, 1995). Cells in a slug move with periodic changes in velocity akin to the chemotactic movements of aggregation (Siegert and Weijer, 1992). Although disrupting the cAMP signal by overexpression or inhibition of the secreted phosphodiesterase deranges development (Traynor *et al.*, 1992; Wu *et al.*, 1995), cAMP does not provide positional information to the developing cells (Wang and Kuspa, 1997).

Precursors to the terminal spores and stalk cells can be identified within a developing aggregate. In the slug, prestalk cells populate the anterior 15–20%, including the tip, and in addition a number of prestalk cells are scattered throughout the posterior of the slug; they are called the anterior-like cells (ALC). There is also a population of prestalk cells at the rear of the slug. Prespore cells constitute the rest of the structure. The two cell types can be distinguished based on a number of criteria, including differential

¹ Current address: Department of Biochemistry, Baylor College of Medicine, 1 Baylor Plaza, Houston, TX 77030.

staining by vital dyes (Bonner, 1952) and expression of tissue-specific marker genes (Williams and Morrison, 1994). Prestalk and prespore cells can also be distinguished based on their differential mobility toward cAMP, either in segregated cells (Mee *et al.*, 1986) or in aggregates (Traynor *et al.*, 1992).

Cell-surface receptors (cARs) bind cAMP and transduce the signal to different cellular responses, including cell movement and gene expression (Hereld and Devreotes, 1992). Permanently elevated levels of extracellular cAMP, which lead to receptor desensitization, are prevented through the action of two extracellular cyclic nucleotide phosphodiesterase (PDE) activities: a secreted soluble form (ePDE) and a membrane-bound form (mPDE). While ePDE levels accumulate as development progresses, mPDE levels accumulate only during aggregation; after the mound forms, mPDE levels fall rapidly. The ePDE can be inactivated by a specific glycoprotein inhibitor (PDI), while the mPDE is immune to inhibition as long as it is bound to the membrane. ePDE and mPDE activities are different forms of the same protein, and all of the detectable mPDE activity has been localized to the outer surface of the cell (Shapiro *et al.*, 1983; Malchow *et al.*, 1972).

The PDE gene (*pdsA*) is developmentally regulated: a vegetative growth-specific promoter is inactivated when development begins, and an aggregation-stage promoter drives expression during early development (Faure *et al.*, 1990). Later in development, a prestalk-specific promoter directs expression at the tip of the slug and in other presumptive stalk cells (Hall *et al.*, 1993), most notably in the prestalk A compartment (Jermyn *et al.*, 1989). Overexpression of the ePDE-specific inhibitor glycoprotein from the prestalk promoter of *pdsA* deranges terminal development (Wu *et al.*, 1995); therefore the presence of PDE during later development is critical.

We have constructed targeted gene disruption mutants of *pdsA* and also modified the gene so that the new mutants can be efficiently complemented. The gene disruption mutants of *pdsA* reflected the phenotype of the previously reported mutants: development was aborted, but the addition of purified ePDE to PDE-null cells restored aggregation competence to the gene disruption mutant (Barra *et al.*, 1980; Faure *et al.*, 1989). This suggested that the PDE protein need only be present outside the cell and serves in the intercellular signaling mechanisms. Thus, a minority of PDE-null cells in a population of wild-type cells should be able to synergize and participate in forming the chimeric structures—but this is not the case. We have used the isogenic null and parental cells to show that, even in the presence of an excess of wild-type cells, the PDE-null cells have a different fate. They move randomly within aggregates. The wild-type cells are gathered in the center of the aggregate, leaving the PDE-null cells more randomly spread, with a large number in the periphery. The null cells fail to become spores at the expected frequency, and become preferentially cells at the front and rear of the slug. The PDE-null cells later populate the upper and lower cups of the sorus, indicat-

ing that they have also formed a portion of the anterior-like cells of the slug. Endogenous production of PDE by each cell is necessary for efficient postaggregation localization and for the formation of slugs. We found that during aggregation the elaboration of streams and large territories depends critically on the amount of PDE provided by wild-type cells. Because the product of *pdsA* is secreted, we attribute this to cAMP receptor desensitization.

MATERIALS AND METHODS

Cell lines. The parental strain DH1 (Caterina *et al.*, 1994) was a gift from P. Devreotes. It is a derivative of the axenic strain Ax3K in which *pyr5-6* has been deleted and requires exogenous uracil for growth. The PDE-null strain uk7 was generated from DH1 by targeted gene disruption, with *pyr5-6* as a selectable marker.

Gene disruptions. pIX1.4 contains the cDNA of *pdsA* under the control of the bacterial *lacZ* promoter and was made by subcloning the 1.4-kb *Bst*BI/*Eco*RI fragment of *pdsA* cDNA (Lacombe *et al.*, 1986) into pUC19 digested with *Acc*I and *Eco*RI. To generate the PDE-null cells, the plasmid pIX1.4 was digested with *Cla*I. This linearized the plasmid at the unmethylated *Cla*I site in the coding sequence of *pdsA*. The plasmid pJB-1, a gift from P. Devreotes, carries a fragment of the *D. discoideum* genome containing *pyr5-6* (Jacquet *et al.*, 1988) in the *Cla*I site of pBluescript II KS (Stratagene). The *pyr5-6* cassette was isolated from pJB-1 by digesting the plasmid with *Cla*I and purifying the 3.7-kb fragment. The fragment was ligated to the *Cla*I-linearized pIX1.4 to generate pUrK⁻. In this plasmid, the direction of transcription of *pyr5-6* was oriented opposite to that of *pdsA*. The plasmid pUrK⁻ was introduced into DH1 cells via electroporation (Howard *et al.*, 1988). Parental cells were grown to a density between 1×10^6 and 3×10^6 cells/ml. The cells were harvested, washed once in buffer (250 mM sucrose, 1 mM sodium phosphate, pH 6.1), and resuspended to a density of 4×10^7 cells/ml in the same buffer. An aliquot of 0.4 ml was electroporated in a 0.2-cm cuvette at 1.0–1.5 kV with 200 Ω at 3 μ F. The time constant was 0.5–0.6 ms. Cells were permitted to recover overnight in HL5 supplemented with uracil. To facilitate isolating independent transformation events, the cells from one electroporation cuvette were diluted in 20 ml FM medium (Franke and Kessin, 1977) and subdivided among the wells of a 96-well microtiter plate for growth under selective conditions. Each cuvette did not yield more than 30 growing wells at a time. The resulting transformants were screened for the expected genotype; of the transformants with the correct genotype, one isolate named uk7 was chosen for all further study.

Development. For multicellular development, cells were washed once in cold Sor+Ca buffer (16.7 mM Na₂H/KH₂PO₄, pH 6.0, supplemented with 50 μ M CaCl₂). The cells were resuspended in Sor+Ca buffer to a density of 1×10^8 cells/ml, and 0.5 ml was plated on Millipore filters resting on Whatman 17 filter pads saturated with Sor+Ca buffer as previously described (Franke *et al.*, 1987) or on 1% agar dissolved in KK2 buffer (20 mM potassium phosphate, pH 6.8), at a density of 3.9×10^5 cells/cm².

Enzyme assays. The PDE activity of the cells was measured in two fractions: the extracellular medium and the cell membranes. Extracellular buffer was harvested by squeezing the filter pads on which the cells developed. To inactivate the PDI, the harvested buffer was dialyzed against 1 mM dithiothreitol (DTT) in 50 mM Tris (50 mM Tris-HCl, pH 7.5) overnight at 4°C, then against 0.1

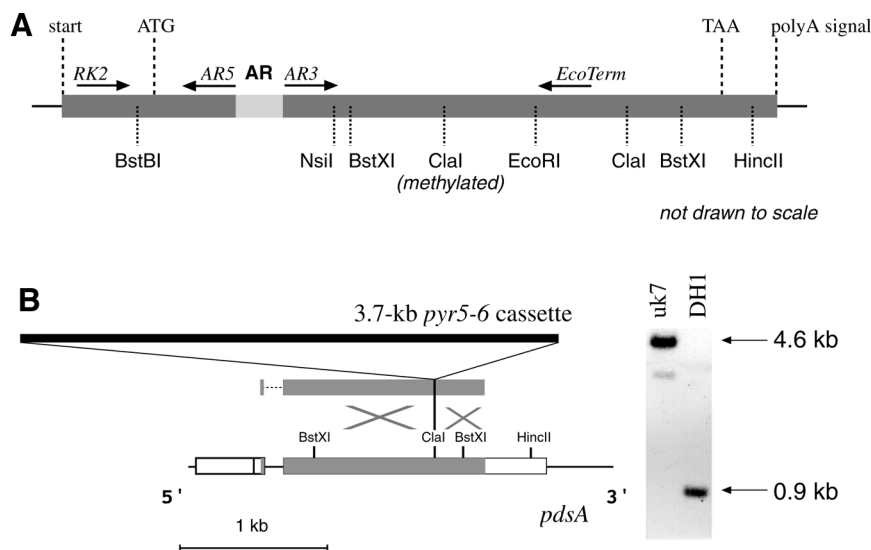


FIG. 1. Strategy used in inactivating *pdsA*. (A) The structure of the gene with relevant restriction sites and primers. All Southern blots employed genomic DNA digested with *Bst*XI and were probed with a 0.9-kb *Bst*XI fragment that contains most of the *pdsA* coding sequence. As expected, a 0.9-kb band is detected in parental DNA digested with *Bst*XI. (B) Disruption in the uk7 strain (uracil selection). See Materials and Methods for details of constructs.

mM DTT in 50 mM Tris for 2–6 h at 4°C. The dialyzed buffer was referred to as the E fraction and contained the ePDE activity. The cells were harvested by vigorous washing of the filters. The harvested cells were resuspended in 0.5 ml of 50 mM Tris, and broken by one round of freezing at –20°C and thawing. Microscopic inspection was used to determine that the majority of cells were lysed using this method. The lysate was centrifuged in a microfuge at 4°C for 30 min to pellet the membrane fraction. The membrane pellet was washed twice with 50 mM Tris. The washes did not contain significant levels of PDE activity. The pellet was dissolved in 0.2% Triton X-100 in 50 mM Tris. Residual debris was pelleted by centrifugation in a microfuge for 15 min. The supernatant from the Triton-extracted membranes contained the mPDE activity. PDE activity was measured using a modification of a previously described radiometric assay (Bulgakov and Van Haastert, 1983).

Deletion of a region of acidic amino acids. A pair of PCR products was used to generate a deletion of the acidic region (AR) of the PDE. A map is shown in Fig. 1A. The primers AR5 (AAGTTA-ACGGAATTTACAATATTTA) and AR3 (AAGTTAACAGTA-ACGATGGTAGTAA) each contain a synthetic *Hpa*I site and match the sequences that flank the acidic region of *pdsA*. The primer EcoTerm (GGTTTAACCTTTGAC) is downstream of the *Eco*RI site in the coding region of *pdsA*, and the primer RK2 (CCC-ACAAACGCCACACACTCAC) is located in the 5' untranslated region of *pdsA*. Genomic DNA from DH1 was employed as a template to amplify a 0.5-kb fragment from the primers RK2 and AR5. This fragment was treated with Klenow enzyme and subcloned into the *Sma*I site of pUC18 to yield p213. The *Hpa*I site was adjacent to the *Bam*HI site of pUC18. The inserted fragment was sequenced to confirm that no PCR errors had occurred between the *Hpa*I site and the internal *Bst*BI site. A 0.7-kb fragment was obtained from the primers AR3 and EcoTerm. This fragment was digested with *Eco*RI and *Hpa*I, and was subcloned into pUC18 that had been digested with *Eco*RI and *Hinc*II to yield the plasmid p301.

The subcloned fragment was sequenced to confirm that no errors had occurred between the synthetic *Hpa*I site and the internal *Nsi*I site. The plasmid p301 was digested with *Hinc*II and *Msc*I, and the resulting 0.6-kb fragment was subcloned into p213 that had been digested with *Hpa*I and *Hinc*II. The resulting plasmids were checked for the presence of an *Hpa*I site; one plasmid, p408, was chosen. The fusion of the synthetic *Hpa*I sites removes all nucleotides that code for the AR and replaces them with a single GTT codon. To minimize the possibility of a PCR error causing an unintended mutation, p408 was digested with *Bst*BI and *Nsi*I to excise the fully sequenced portion of the fused PCR products, and this fragment was subcloned into pVH (pUC18 containing the *Eco*RV to *Hinc*II fragment of the *pdsA* genomic clone) that had also been digested with *Bst*BI and *Nsi*I. The result was pVHΔAR. To attach the appropriate promoters to the construct, the plasmid pGP-1, which is a genomic clone of *pdsA* with its vegetative and late promoters carried in pBR322 (Faure *et al.*, 1989), was digested with *Hind*III and *Aat*II, treated with Klenow, and religated. This eliminated a pBR322-borne *Eco*RI site and rendered the *Eco*RI in the *pdsA* coding sequence unique. The modified plasmid was renamed pGP-1.1 to reflect this change. The plasmid pVHΔAR was digested with *Bst*BI and *Eco*RI to excise the 0.9-kb fragment that lacks the AR, and this fragment was subcloned into pGP-1.1 that had also been digested with *Bst*BI and *Eco*RI. The resulting plasmid, pGP-1.1ΔAR, contains *pdsA* without the AR, under the control of the vegetative and late promoters. To generate a construct of the modified gene (*pdsA*ΔAR) under the control of the aggregation promoter, the plasmid pGP-1.1ΔAR was digested with *Nru*I and *Bst*BI to remove the vegetative and late promoters. The aggregation promoter was excised from the plasmid pAP53 (Hall *et al.*, 1993) with *Sma*I and *Acc*I and this fragment was cloned into pGP-1.1ΔAR. The new plasmid, pGPAΔAR, contains the modified *pdsA* under the control of the aggregation promoter.

Dot blots. To assay for changes in specific activity, the ePDE

extracts prepared from the different transformant strains were diluted to the same concentration of PDE units/ml, and serial dilutions were immobilized on nitrocellulose using a dot blot assay. The blots were probed with polyclonal antibodies raised against a synthetic peptide corresponding to the N-terminus of the processed protein (Franke *et al.*, 1991) and developed using horseradish peroxidase-conjugated secondary antibodies and the Renaissance chemiluminescence kit (NEN/Dupont).

Transformants. The constructs were cotransformed with pB10TP-2 into PDE-null cell line uk7. The plasmid pB10TP-2 carries a *neo^R* cassette that confers resistance to the drug G418 (Early and Williams, 1987). The resulting transformed cells were screened by Southern blot for multiple-copy integration events. Northern blots were prepared from confirmed transformants using standard techniques to verify overexpression of the transformed plasmid.

The strain uk7-A15gal was constructed by transforming uk7 with the plasmid pA15gal (Shaalsky *et al.*, 1995), which carries the *neo^R* marker and the bacterial β -galactosidase gene *lacZ* under the control of the *D. discoideum* actin-15 promoter. All cells used in the mixing experiments were isogenic derivatives.

Tracking cells. To track cell movements in chimeric mixtures, a representative number of cells were labeled with a fluorescent dye prior to mixing. Cells were labeled with rhodamine dextran, CellTracker green, or CellTracker red (Molecular Probes). To label *D. discoideum* cells with CellTracker, washed cells were kept in suspension with KK2 buffer containing 50 μ M CellTracker red or green at a density of 1×10^7 cells/ml for 30–60 minutes, and then washed once in KK2 prior to use. All cells were stained by this method, and they remained viable. CellTracker red yielded a punctate staining pattern, while CellTracker green had a more uniform cytosolic stain. All gave the same results. The best staining was obtained with CellTracker green, which was used in subsequent tracking experiments. Fluorescently labeled cells constituted 2% of the total number of cells in a chimeric mixture.

Dark-field waves and single-cell tracks were recorded and analyzed using equipment and methods previously reported (Siebert and Weijer, 1989, 1992, 1995; Siebert *et al.*, 1994). Briefly, structures were imaged with a standard inverted microscope using a CCD camera; digital images were averaged at regular intervals (usually every 20 s), and stored onto a laser disk, under the control of an Intel standard 486-based computer equipped with a custom-built video capture board. Analyses of images were performed using custom written software (Siebert and Weijer, 1995).

Participation in spore formation. Since the terminally differentiated stalk cells are inviable, participation in spore formation can be measured by permitting spores from chimeric fruiting bodies to germinate and scoring for mutant clones that appear. PDE-null cells are unable to aggregate, and form easily detected flat plaques on a lawn of bacteria. PDE-null (uk7) cells and parental (DH1) cells were mixed, and an aliquot was plated on *Klebsiella aerogenes* lawns to determine the ratio of flat plaques to aggregating clones. The remainder of the mixture was placed on sterile KK2 agar and permitted to develop under conditions described earlier. After 30 h, when fruiting bodies are evident, the plate was washed with cold KK2 buffer, and the harvested cells were washed once. The spores were diluted to 2×10^6 /ml and incubated at 45°C for 30 min, allowed to cool for 30 min, and heated to 45°C for another 30 min. This ensures killing of cells that have not differentiated into spores (Cotter and Raper, 1968); because heating kills some wild-type cells, we also killed nonspore cells with detergent (0.1% Triton X-100). After heat or detergent treatment, spores were counted and plated onto *K. aerogenes* lawns.

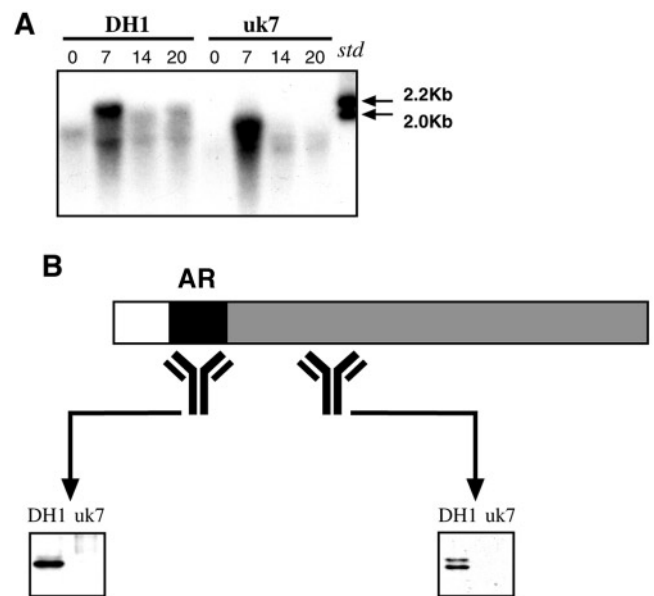


FIG. 2. Northern and Western blots show an altered mRNA and no cross-reacting material in the mutants. (A) Northern blots of gene disruptant uk7, alongside its isogenic parent DH1. RNA was isolated from different stages of development, and the blots were probed with labeled 0.9-kb *Bst*XI fragment. The hours of development on filters are indicated on top of the lanes. (B) Western blots with antibodies raised against the AR epitope (left) or the main peptide of the PDE (right). The supernatants were harvested from cells that had been starved for 6 h in Sor+Ca buffer.

***LacZ* and neutral red staining.** In chimeric mixtures involving cells that express *lacZ*, cells were plated on circles of Whatman No. 50 filter paper saturated with KK2 buffer. At the appropriate stages, filters were fixed in 1% glutaraldehyde, and the presence of β -galactosidase was histochemically detected as previously described (Hall *et al.*, 1993). Neutral red indicates the presence of acidic autophagic vacuoles and was first used by Bonner (1952).

RESULTS

Gene disruption. The PDE-null strain uk7 was generated from DH1 by targeted gene disruption with *pyr5-6*, utilizing a double crossover strategy (Fig. 1B). Northern blots of RNA isolated from uk7 detected a residual mRNA species that hybridized with the PDE probe (Fig. 2A). The detected message was truncated by 0.2–0.3 kb, consistent with the predicted messenger RNA that would result from the gene disruption. The developmental timing was normal.

Enzyme assays detected no residual PDE activity in the mPDE and ePDE fractions of the *pdsA* disruptant (data not shown). Western blots, using two independent antibodies that recognize sequences in the amino-terminus of the PDE, failed to detect any protein in the disruptant (Fig. 2B). The first antibody was raised against the region of 27 acidic

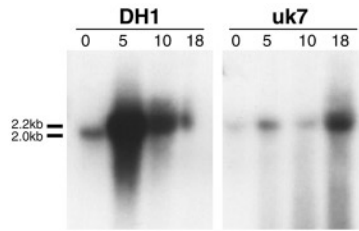


FIG. 3. Absence of PDE suppresses expression of the *car1* transcript. A Northern blot of a developmental time course of parental and mutant strains is shown. Numbers indicate time in hours after the onset of starvation.

amino acids (AR) in the amino-terminus and only recognizes the full-length PDE. The second antibody was raised against a peptide sequence in the body of the secreted protein (Franke *et al.*, 1991) and recognizes both the full-length protein and the truncated form that appears *in vivo* when the AR is removed. Because of the absence of enzymatic activity and cross-reacting material, we considered these cells PDE-null cells. Similar results were obtained with the PDE-null strain called PEN, which was created by introducing a plasmid bearing the *neo*^R cassette into the downstream *Bst*XI site of the *pdsA* locus of strain AX3K, employing a single crossover strategy (data not shown).

The PDE-null cells do not enter multicellular development. This is consistent with the predicted phenotype based on the original *pdsA* mutants (Barra *et al.*, 1980; Faure *et al.*, 1989). When purified PDE was added to a lawn of PDE-null cells, the mutant cells became competent to aggregate, although they did not proceed beyond the loose aggregate stage. The inability of PDE-null cells to hydrolyze extracellular cAMP may result in constitutively elevated extracellular cAMP levels. The high-affinity cAMP receptor *car1* is the primary transducer of the extracellular cAMP signal, but it desensitizes when cAMP levels remain high. Thus, the *car1* receptors may be inactive in PDE-null cells. The *car1* receptor is encoded by *carA*, which is regulated by aggregation and late stage promoters (Saxe *et al.*, 1991). The *carA* aggregation promoter is sensitive to levels of extracellular cAMP: it is strongly induced by oscillating levels of cAMP, but is down-regulated by constant levels of extracellular cAMP. The late promoter of *carA* is induced by high steady-state levels of extracellular cAMP. Northern blots of RNA derived from the PDE-null cells show that, unlike the parental cells, the PDE-null cells do not exhibit high levels of *carA* message at 5–10 h of starvation, although a later stage message is expressed (Fig. 3). The expression pattern of the *carA* transcript in the PDE-null cells is consistent with exposure to steady-state levels of extracellular cAMP.

Complementation of the null mutant. When a plasmid containing the unmodified coding sequence for PDE was

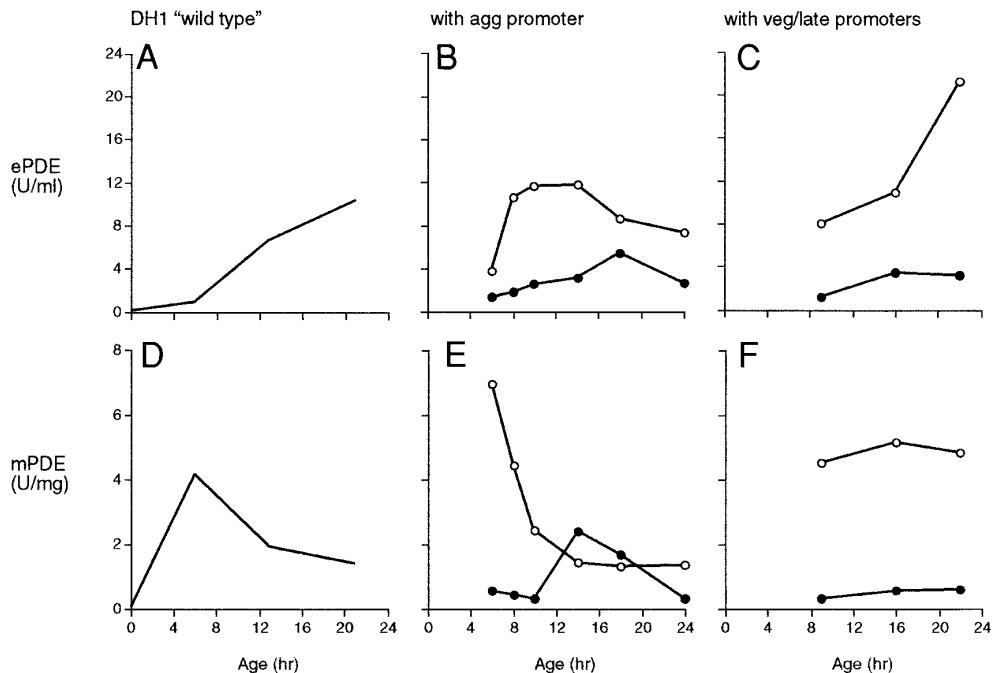


FIG. 4. Higher levels of PDE activity result from constructs that lack the acidic region. (A and D) The normal expression pattern of extracellular and membrane-bound PDE activity, respectively. (B and E) The patterns when the entire *pdsA* is expressed from the aggregation-specific promoter. (C and F) PDE activities when the PDE is driven by the prestalk-specific late promoter of the PDE. In all panels the curves with open circles represent *uk7* transformed with *pdsA*ΔAR; closed circles represent *uk7* transformed with *pdsA*.

introduced into PDE-null cells, aggregation and a degree of development were restored to the mutant cells only when the cells were plated at high densities. Under low-density monolayer conditions (see Materials and Methods), the transformants did not aggregate, regardless of the promoters controlling the introduced copies of *pdsA*. Transformants were maintained under stringent selection conditions to guarantee high copy numbers of the integrated *pdsA*, and presumably high expression levels of the protein. In addition, the isolates from each transformation were screened by Southern blot to confirm that the introduced *pdsA* was present in multiple copies. When the transformants were assayed for PDE activity, they expressed less than wild-type ePDE activity and less mPDE activity (Fig. 4). Northern blots of the transformants demonstrated that the introduced copies of *pdsA* made messenger RNA of the correct size (data not shown).

A form of the PDE from which the AR has been cleaved is found exclusively in the secreted fraction. The 27 amino acids of the AR were deleted to test the hypothesis that it is involved in anchoring the PDE to the membrane. While this was not the case, we did find that, when constructs containing *pdsA* with the Δ AR modification were used to transform PDE-null cells, the transformant cells synthesized higher levels of ePDE and mPDE activities (Fig. 4). The AR is not needed for membrane attachment of the enzyme, but appears to be an inhibitory domain of the PDE protein. This inhibition is separate from that which occurs by interaction of the PDE and the PDI.

The low PDE activity measured from uk7 transformants overexpressing *pdsA* mRNA may be caused by the inability to translate the messenger RNA or the synthesis of a protein that is less active. These two possibilities were distinguished by comparing the relative amounts of PDE protein in extracts containing identical amounts of PDE activity. Aliquots of supernatants from transformants of uk7 that contain equal units of PDE activity were immobilized on nitrocellulose as dot blots and detected using a PDE antibody. The supernatants from transformants with the Δ AR construct under control of the aggregation promoter exhibited a signal similar to the parental DH1 strain, while constructs with an intact PDE coding sequence showed more

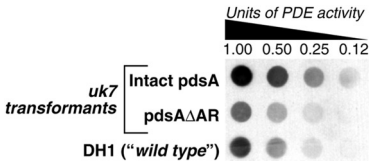


FIG. 5. Dot blot showing that transformants expressing the Δ AR construct have higher specific activity PDE. Cells were starved in shaking suspension for 8 h, and the supernatant was harvested, dialyzed against DTT to inactivate the PDI, and carefully assayed to determine the PDE activity. Equal units of enzyme extract were immobilized on nitrocellulose membrane, and the PDE was detected using the antibody raised against the main peptide (Fig. 2B).

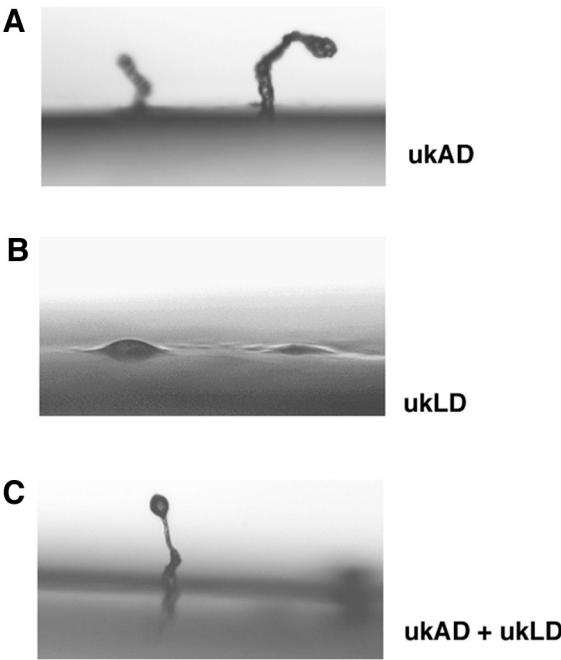


FIG. 6. Complementation of the PDE-null aggregate-less phenotype. (A) The phenotype of uk7 cells overexpressing *pdsA* Δ AR from the aggregation promoter. (B) The phenotype when expression is from the vegetative/late promoters. If the two transformants are mixed in a 1:1 ratio, morphologically normal fruiting bodies are restored (C).

cross-reacting material (Fig. 5). This was also true if the vegetative/late promoters were in control of the gene (not shown). Thus, removing the 27 amino acids of the acidic region increases the specific activity of the enzyme. This removal also occurs during the normal process of enzyme maturation *in vivo*, probably extracellularly.

Unlike the transformants of uk7 with the complete PDE coding sequence, transformants carrying multiple copies of *pdsA* Δ AR were able to aggregate under the low-density monolayer conditions, consistent with the ability of the Δ AR modification to make more functional PDE. The terminal phenotypes of the *pdsA* Δ AR transformants depended on the promoters that controlled the gene. If the *pdsA* Δ AR is under the control of the aggregation promoter, the transformants make stalk-like structures (Fig. 6A). Under the control of the vegetative and late promoters, the *pdsA* Δ AR transformants stopped at the mound stage (Fig. 6B). This is the same phenotype observed when the *pdsA* is overexpressed in wild-type cells under the control of the vegetative/late promoters, and is thought to be a consequence of reducing the cAMP concentration within the mound (Traynor *et al.*, 1992). If the two different transformants (PDE-nulls expressing *pdsA* Δ AR from the aggregation promoter only and PDE-nulls expressing *pdsA* Δ AR from the vegetative and late promoters only) were mixed in a 1:1

ratio, morphologically normal fruiting bodies resulted (Fig. 6C). Both the aggregation and the late promoters are necessary for normal development, although they do not have to reside in the same cells to produce a phenotypically normal fruiting body. The two types of transformant may have different fates within the apparently normal fruiting body.

The form of aggregation, the behavior of cells in mounds, and the formation of slugs depend on the level of PDE. A small proportion of parental cells can restore development to PDE-null cells, but at low ratios of wild-type cells, the normal aggregation streams are not found. When PDE-null cells were mixed with parental cells in a ratio of 9:1, no

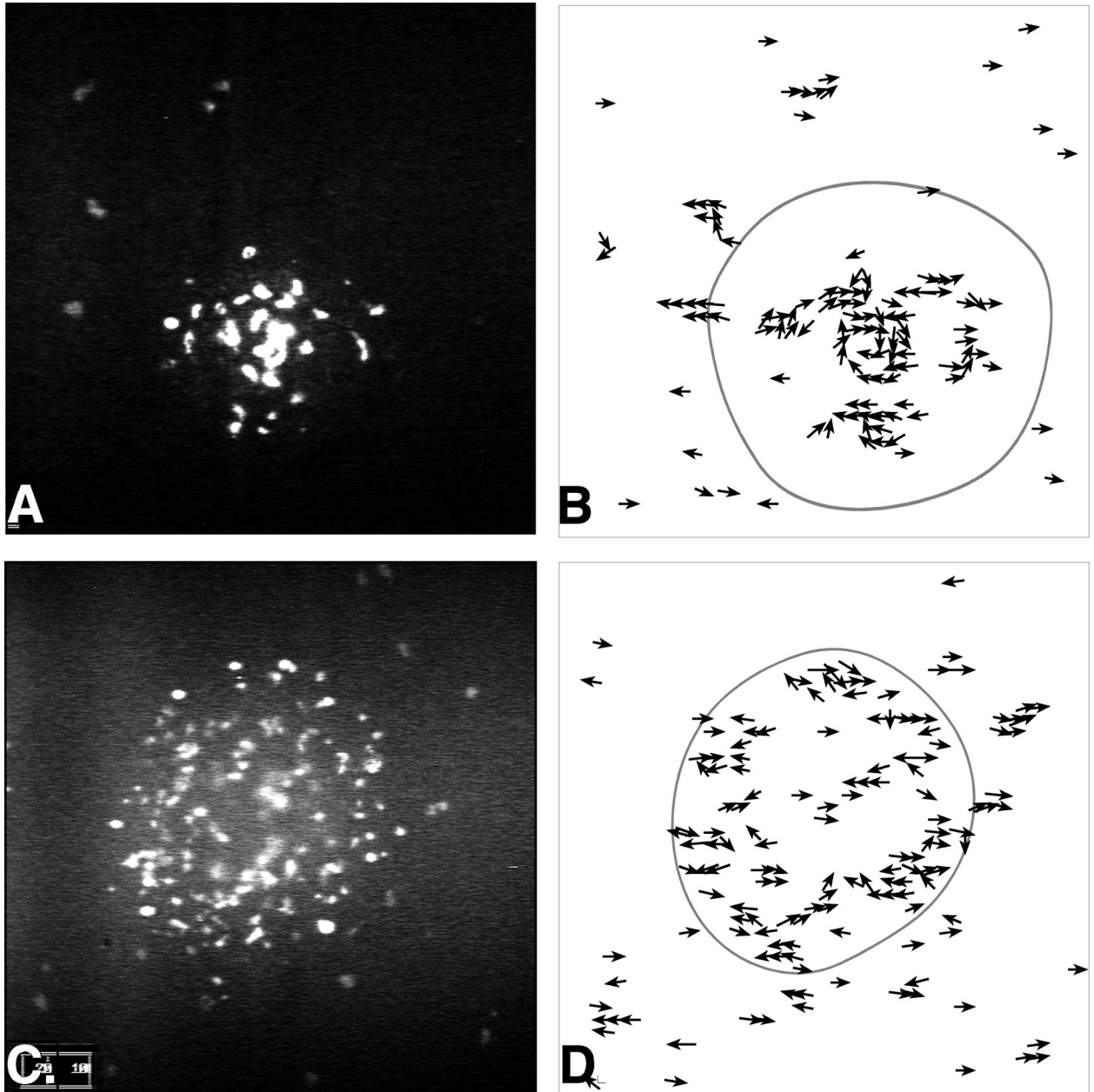


FIG. 7. PDE-null cells move randomly in chimeric mounds. The PDE-null mutant strain uk7 was mixed with the parental strain DH1 at a ratio of 9:1 (90% mutant to 10% wild type) and allowed to develop on agar plates. Prior to mixing, a small number of cells from either strain were labeled with CellTracker green. (A and C) Fluorescence images of early mounds generated from these mixtures. (B and D) Vector plots showing the direction of the movement of the labeled cells; the gray line denotes the boundary of the mound. (A) is from a mound with the parental DH1 cells labeled. In C, mutant cells were labeled.

streams of migrating cells were observed, but aggregates still formed as single cells made their way directly into centers. The mounds that formed from these aggregates were smaller than those from 100% DH1 cells. Thus 10% of the normal PDE is insufficient to gather cells from a large territory. When the concentration of isogenic wild-type cells passes 20%, normal aggregation is progressively restored. We speculate that the elaborate regulation of *pdsA* has evolved to gather cells from a large area.

In early mounds generated from the chimeric mixtures containing 10% DH1 cells, the optical density waves propagate as concentric rings from the center to the periphery, unlike the multiarmed spirals of the parental DH1 cells (R. Succgang and C. Weijer, unpublished observations). No slugs formed from these mounds, which proceeded directly to culmination. This result suggests that slug formation requires substantial expression of PDE, and is consistent with the results of Wu *et al.* (1995), who showed that suppressing PDE by overexpressing PDI from a late-stage PDE promoter also blocked slug formation.

Parental cells segregate from mutant cells in chimeric aggregates. We asked how the behavior of PDE-null cells in mounds differs from that of isogenic parental cells. Each cell population was labeled with a fluorescent dye, and time-lapse videomicroscopy was used to capture the movements of the labeled population as development proceeded. When 2% of the cells in a DH1 population (wild type) were fluorescently labeled, the labeled cells migrated into mounds and entered a characteristic spiral motion, with periodic changes in velocity similar to those observed during chemotactic aggregation (Siegert and Weijer, 1989). In a chimeric mound formed from a mixture of 10% DH1 cells (of which 20% were labeled) and 90% PDE-null cells, the wild-type cells have the directed motions characteristic of a mound made of 100% wild-type cells (Figs. 7A and 7B). In contrast, the mutant cells, which comigrated with parental cells during aggregation, did not move in a coordinated rotational manner in the mound (Figs. 7C and 7D) and instead moved randomly. The images in Figs. 7A and 7C are single frames in a series of captured images. The arrows in the vector plots in Figs. 7B and 7D show the movement of the cells over 16 frames, captured 20 s apart. Parental DH1 cells segregated from the mutant cells by moving into the center of the mound (Figs. 7A and 8).

If the parental cells constituted the majority of the cells (2% PDE-null cells and 98% parental cells), early macroscopic aggregation phenomena such as dark-field waves and stream formation occurred, PDE-null cells still moved randomly in the mounds, and the wild-type cells were drawn by rotating waves into the center (data not shown), as was the case with the experiments shown in Figs. 7 and 8.

As development proceeded, some PDE-null cells were recruited to the nascent tip. This is reflected in the finger and slug stages, where the labeled PDE-null cells populated the rear and tip regions of the slug (Figs. 9A and 9B). Figure 9C shows the PDE-null cells in the stalk and lower cup of chimeras. The lower cup is formed by the ALC population (Williams and Morrison, 1994). Figure 9F shows the sorting

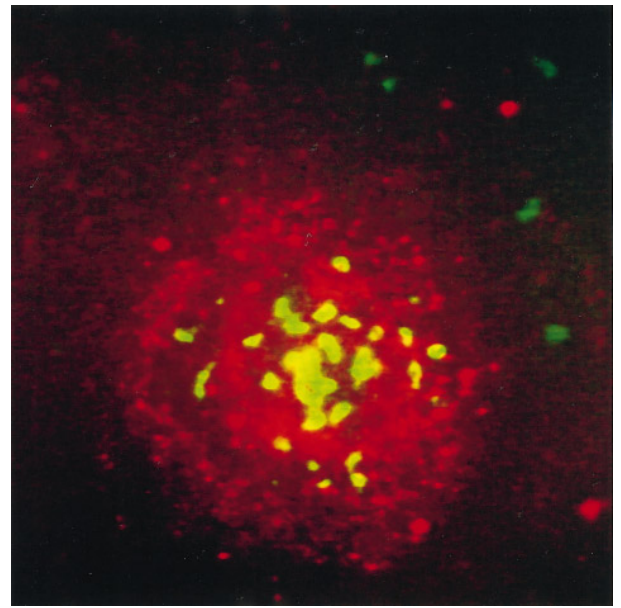


FIG. 8. Segregation of PDE-null and parental cells. Images of a single mound formed from 90% PDE-null cells and 10% parental cells, with subpopulations of each strain labeled with CellTracker green (parental) or CellTracker red (PDE-null cells). Fluorescently labeled cells of each kind constitute 2% of the total cell population. The initial gray scale images were pseudocolored to distinguish the double label and the two images were superimposed.

of PDE-null cells to the papilla of the fruiting body, which is a prestalk region. Figure 9G shows cells that have been complemented with the modified *pdsA* described above; they no longer localize to the papilla, but populate the spore mass. When the parental cells in the chimeras were marked with CellTracker green, to locate the terminal fate of the wild-type cells, we found that they preferentially populated the spore mass (Figs. 9D and 9E).

Segregation of PDE-null cells is also shown by PDE-null cells constitutively expressing the *lacZ* marker from a non-cell-type-specific actin promoter (Fig. 10). Labeled wild-type cells distributed randomly throughout the aggregate and participated in both stalk and spore formation (data not shown). When *lacZ*-expressing PDE-null cells made up 20% of the population, they preferentially populated the rear of the slug (Fig. 10A). At higher proportions (80% uk7-A15gal:20% Ax3K), mutant cells could also be observed in the tip of the slug (Fig. 10B), although slugs form less frequently due to the reduced PDE. Mutant cells were observed throughout the stalk and the lower cup of the sorus (Fig. 10C). The uk7 cells were also labeled with neutral red, which stains an acidic autophagic vacuole that disappears in prespore cells (Fig. 11). In chimeras of 50% uk7 and 50% AX3, the stained cells were in the anterior compartment and in the rear of the slug, and in fruiting bodies the stained uk7 cells labeled the base of the fruiting body, the lower and upper cups, and the papilla. These

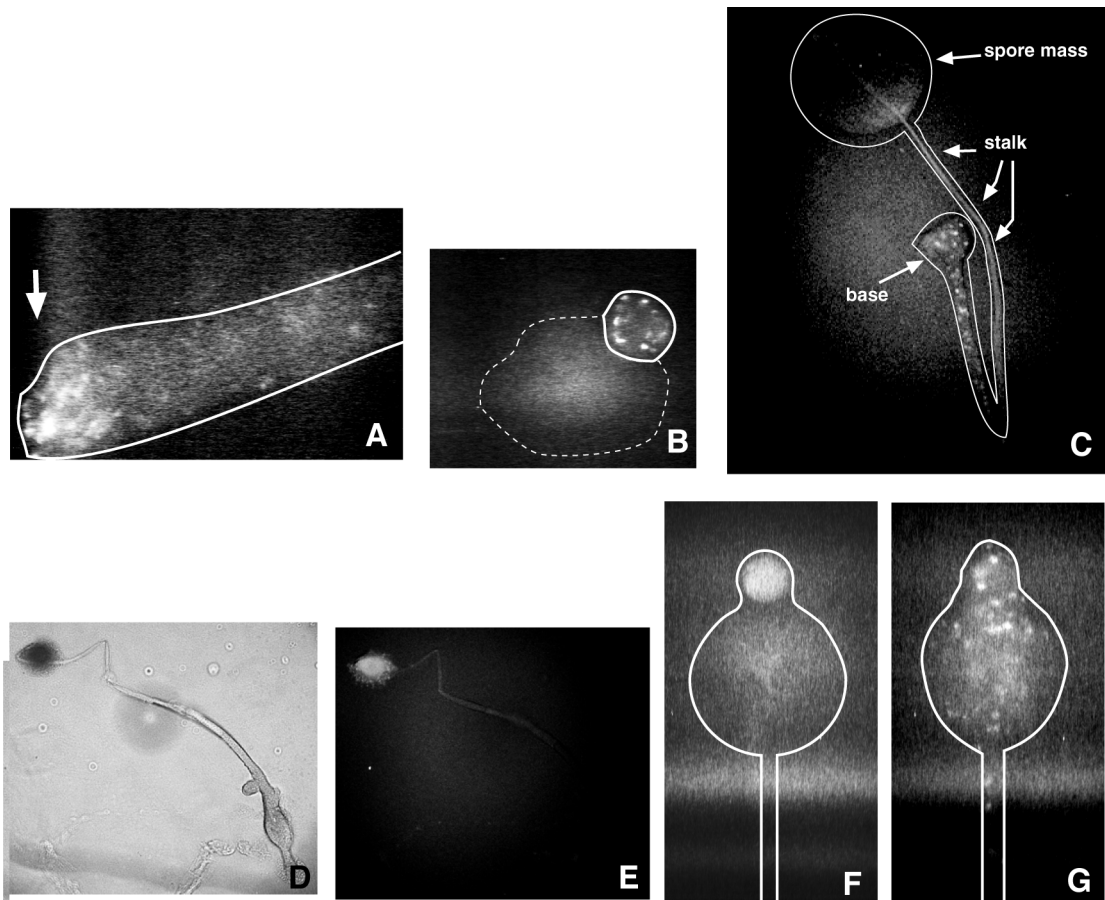


FIG. 9. Behavior of PDE-null cells in postaggregation stages. PDE-null cells were labeled with the fluorescent dye CellTracker green and mixed at a ratio of 1:50 with parental cells. (A) The PDE-null cells coaggregated with the parental cells, but when the slug formed, labeled cells were found mostly in the rear end of the slug (arrow). (B) An image of a slug oriented toward the viewer. The tip is in focus and is encircled with a solid white line, demonstrating that a number of the fluorescently labeled PDE-null cells were present there. The body of the slug is out of focus, and the border is denoted by the broken line. (C) The chimeric fruiting body. (D) A fruiting body formed from a mixture of 10% parental DH1 cells with 90% PDE-null cells. One-fifth of the parental cells were labeled with CellTracker green. (E) The same fruiting body as in D, showing that the parental DH1 cells preferentially populate the spore mass. (F) A control mixture of 90% uk7 cells, 10% DH1 cells, with 2% of the uk7 cells labeled. The labeled cells are in the papilla. The white line delineates the border of the structures. (G) A chimera formed from DH1 cells and uk7 cells transformed with *pdsAΔAR* under the control of the vegetative/late promoters. Transformant cells were labeled with CellTracker green. During the slug stage, transformants also were found throughout the structure, akin to wild-type cells (not shown).

are the regions that in wild-type strains contain prestalk cells and anterior-like cells.

If a cell in a chimera is at a disadvantage, this should be reflected in the less efficient formation of spores. The spores from chimeric mixtures exhibited far fewer PDE-null cells than the starting population (Table 1), indicating that PDE-null cells are less likely to differentiate into spores, even if surrounded by a fivefold excess of parental cells producing PDE.

DISCUSSION

Defined mutations in critical genes offer a way to compare mutants with isogenic parents. We have disrupted the

gene that codes for the cyclic nucleotide phosphodiesterase, which controls extracellular cAMP levels and hence the chemotactic responsiveness of cells at several stages of development. Gene disruption mutants made an altered PDE mRNA, consistent with the localization of the insert in the 3' end of the gene, but no gene product could be detected by enzyme assay or by Western blots.

In the *Dictyostelium* PDE a stretch of 27 acidic amino acids (AR) follows the signal peptide; this region is normally removed in the enzyme that is secreted into the medium. Only with constructs which lack the AR could complete complementation be achieved, allowing cells to aggregate and develop at low cell densities. We postulate that this is

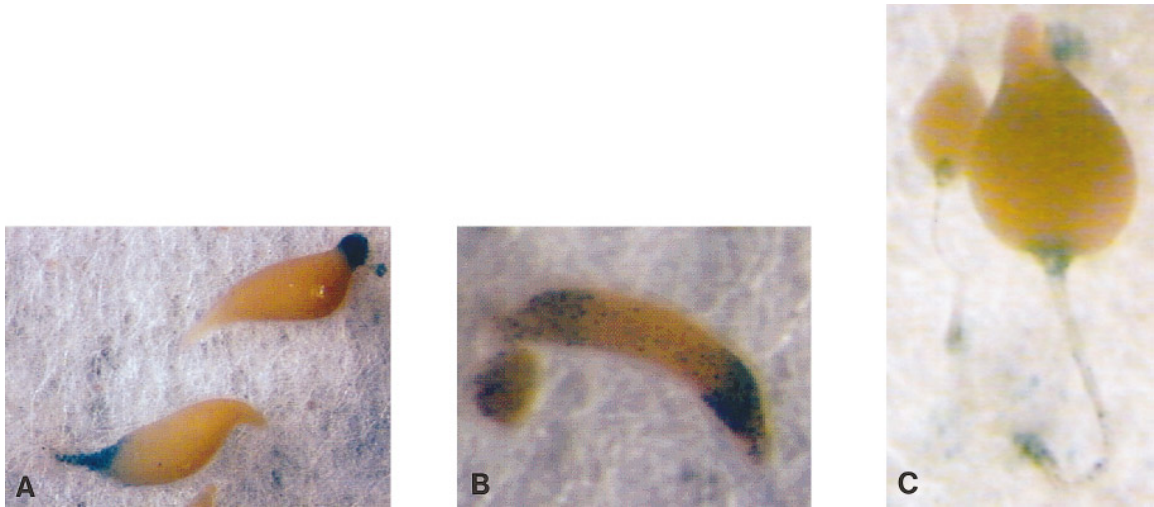


FIG. 10. PDE-null cells preferentially populate prestalk and stalk regions in chimeras. The PDE-null strain uk7 was marked by introducing a constitutively expressed copy of the bacterial *lacZ*. (A) A pair of slugs formed from a mixture of 10% uk7 with 90% Ax3K; blue cells are seen almost exclusively in the rear-guard population. If the proportion of PDE-null cells is increased to 80% of the total population, the incidence of slug formation decreases, but slugs that form have the PDE-null cells present in the tip of the slug as well (B). Later, X-gal-stained cells appear in the stalk and lower cup of the fruiting body (C).

due to the higher enzyme specific activity. This conclusion is based on enzyme assays during time courses of development. Normalizing the results of enzyme assays to the amount of cross-reacting material detected in dot blots indicates that the specific activity of the enzyme is increased when it lacks the AR. The complementation studies also

showed that the aggregation-specific and the later prestalk-specific promoters are both necessary for normal development. Complementation with constructs driven by either promoter alone results in an aggregate, but not a complete fruiting body. When the two strains are mixed, development of mature fruiting bodies is restored.



FIG. 11. PDE-null cells in chimeras express prestalk cell-specific structures. Neutral red-labeled uk7 or AX3 cells were mixed with an equal number of unstained AX3 cells. Aggregates were allowed to form on agar, and the dish was wrapped so that only a slit was open to admit light. The slugs formed and migrated toward the light for 36 h before they were photographed.

TABLE 1
PDE-Null Cells in a Chimera Have a Decreased Probability of Forming Spores^a

Method of killing nonspores	Time of scoring	Total no. of clones scored	No. of PDE-null clones	% PDE-null cells in the chimera
Heating	Before development (vegetative cells)	103	17	16.5%
	After development (spores)	699	23	3.3%
Detergent	Before development (vegetative cells)	710	136	19.5%
	After development (spores)	851	11	1.3%

^a Mixtures of PDE-null cells and parental cells were made, and the ratio was determined by plating them on lawns of *K. aerogenes*. The proportion of mutant cells was determined by scoring the number of flat plaques compared to the total number of clones. The mixture was permitted to develop, and the cells were harvested after culmination. Cells that failed to become spores were killed by heat treatment or washing in detergent.

cAMP serves a chemotactic role during aggregation, but the details of its role in later development are still being elaborated. We therefore examined the behavior of PDE-null cells in chimeras with isogenic parental cells. Because PDE is an extracellular enzyme and is exported by a secretory pathway, we expected that mixing a few PDE-null cells with a large number of wild-type cells would compensate for the defect in the null cells. During aggregation, when the cells are relatively separate, this is apparently the case. PDE-null cells move into aggregates in a normal way. Once the cells have converged, in the much closer confines of the aggregate, the PDE-null cells behave differently. In mounds, the parental cells move in concentric circles around signaling centers at the apex of the mound. The PDE-null cells, even when present as a small minority (1 in 50), fail to move in the directed manner of the PDE-expressing cells. We interpret this to mean that the cAMP receptors (either cAR1, which is generally expressed, or cAR2, which is prestalk specific) are not properly deadapted. As shown in Fig. 3, this appears to be the case for the cAR1 transcript during aggregation.

One explanation for the results is that PDE from adjacent cells does not suffice because deadaptation depends on the close association of PDE and cAMP receptors. It has recently been shown that cAR1, adenylate cyclase, and PDE coexist in plasma-membrane microdomains (Xiao and Devreotes, 1997), and it is possible that full signal attenuation is only achieved when PDE exists in this configuration.

There is no reason to believe that the *pdsA* gene product has a cytosolic effect on cAMP as an intracellular signaling molecule. It is a glycosylated enzyme with a leader sequence, which presumably is efficiently sorted to membranous compartments of the secretory pathway and the extracellular space. Recently, an intracellular phosphodiesterase was described (Shaulsky *et al.*, 1996). This gene, called *regA*, produces a PDE that is cAMP specific (G. Shaulsky and W. F. Loomis, pers. comm.).

In chimeras of PDE-null cells and parental cells, the mutant cells fail to follow the circular pattern of the parental cells. The consequence is that eventually the parental cells tend toward the center of the aggregate, and many of the

PDE-null cells are left at the periphery. The next step in development, following the mound stage, is an elongation of the mound to form a structure called a finger or standing slug. This occurs under the control of the centrally located apical tip and pulls up centrally located cells first. The cells on the periphery of the mound are left to form the rear of the slug, as shown in Figs. 9 and 10. A few of the PDE-null cells migrate to the tip and form the papilla, stalk cells, and lower cup. Only a minority of the null cells form spores, a number which can be quantified. If the effect of the mutation had been nil, and PDE from other cells could compensate, we would expect 80% of the PDE-null cells to form spores. Instead, only 6–20% of the expected number of spore cells are made by the PDE nulls (Table 1). The results show that cAMP is a critical signaling molecule in the mound and during later stages and that its degradation is a cell-autonomous event.

ACKNOWLEDGMENTS

The authors gratefully acknowledge Jens Rietdorf, Dirk Dormann, Victor Zaydfudim, and Judy Venuti for their assistance. This work was supported by NIH Grant GM33136 to R.H.K.

REFERENCES

- Barra, J., Barrand, P., Blondelet, M. H., and Brachet, P. (1980). *pdsA*, a gene involved in the production of active phosphodiesterase during starvation of *Dictyostelium discoideum* amoebae. *Mol. Gen. Genet.* **177**, 607–613.
- Bonner, J. T. (1952). The pattern of differentiation in amoeboid slime molds. *Am. Naturalist* **86**, 79–89.
- Bulgakov, R., and Van Haastert, P. J. M. (1983). Isolation and partial characterization of a cyclic GMP-dependent phosphodiesterase from *Dictyostelium discoideum*. *Biochim. Biophys. Acta* **756**, 56–66.
- Caterina, M. J., Milne, J. L. S., and Devreotes, P. N. (1994). Mutation of the third intracellular loop of the cAMP receptor, cAR1, of *Dictyostelium* yields mutants impaired in multiple signaling pathways. *J. Biol. Chem.* **269**, 1523–1532.

- Cotter, D., and Raper, K. (1968). Properties of germinating spores of *Dictyostelium discoideum*. *J. Bacteriol.* **96**, 1680–1689.
- Early, A. E., and Williams, J. G. (1987). Two vectors which facilitate gene manipulation and a simplified transformation procedure for *Dictyostelium discoideum*. *Gene* **59**, 99–106.
- Faure, M., Franke, J., Hall, A. L., Podgorski, G. J., and Kessin, R. H. (1990). The cyclic nucleotide phosphodiesterase gene of *Dictyostelium discoideum* contains 3 promoters specific for growth, aggregation, and late development. *Mol. Cell. Biol.* **10**, 1921–1930.
- Faure, M., Podgorski, G. J., Franke, J., and Kessin, R. H. (1989). Rescue of a *Dictyostelium discoideum* mutant defective in cyclic nucleotide phosphodiesterase. *Dev. Biol.* **131**, 366–372.
- Franke, J., and Kessin, R. (1977). A defined minimal medium for axenic strains of *Dictyostelium discoideum*. *Proc. Natl. Acad. Sci. USA* **74**, 2157–2161.
- Franke, J., Faure, M., Wu, L., Podgorski, G. J., and Kessin, R. H. (1991). Cyclic nucleotide phosphodiesterase of *Dictyostelium discoideum* and its glycoprotein inhibitor: Structure and expression of their genes. *Dev. Genet.* **12**, 104–112.
- Franke, J., Podgorski, G. J., and Kessin, R. H. (1987). The expression of two transcripts of the phosphodiesterase gene during the development of *Dictyostelium discoideum*. *Dev. Biol.* **124**, 504–511.
- Hall, A. L., Franke, J., Faure, M., and Kessin, R. H. (1993). The role of the cyclic nucleotide phosphodiesterase of *Dictyostelium discoideum* during growth, aggregation, and morphogenesis: Overexpression and localization studies with the separate promoters of the *pde*. *Dev. Biol.* **157**, 73–84.
- Hereld, D., and Devreotes, P. N. (1992). The cAMP receptor family of *Dictyostelium*. *Int. Rev. Cytol.* **137B**, 35–47.
- Howard, P. K., Ahern, K. G., and Firtel, R. A. (1988). Establishment of a transient expression system for *Dictyostelium discoideum*. *Nucleic Acids Res.* **16**, 2613–2623.
- Jacquet, M., Guilbaud, R., and Garreau, H. (1988). Sequence analysis of the DdPYR5-6 gene coding for UMP synthase in *Dictyostelium discoideum* and comparison with orotate phosphoribosyl transferases and OMP decarboxylases. *Mol. Gen. Genet.* **211**, 441–445.
- Jermyn, K. A., Duffy, K. T., and Williams, J. G. (1989). A new anatomy of the prestalk zone in *Dictyostelium*. *Nature* **340**, 144–146.
- Lacombe, M. L., Podgorski, G. J., Franke, J., and Kessin, R. H. (1986). Molecular cloning and developmental expression of the cyclic nucleotide phosphodiesterase gene of *Dictyostelium discoideum*. *J. Biol. Chem.* **261**, 16811–16817.
- Malchow, D., Nagele, B., Schwartz, H., and Gerisch, G. (1972). Membrane-bound cyclic AMP phosphodiesterase in chemotactically responding cells of *Dictyostelium discoideum*. *Eur. J. Biochem.* **28**, 136–142.
- Mee, J. D., Tortolo, D. M., and Coukell, M. B. (1986). Chemotaxis-associated properties of separated prestalk and prespore cells of *Dictyostelium discoideum*. *Biochem. Cell Biol.* **64**, 722–732.
- Saxe, C. L., III, Johnson, R. L., Devreotes, P. N., and Kimmel, A. R. (1991). Expression of a cAMP receptor gene of *Dictyostelium* and evidence for a multigene family. *Genes Dev.* **5**, 1–8.
- Shapiro, R. I., Franke, J., Luna, E. J., and Kessin, R. H. (1983). A comparison of the membrane-bound and extracellular cyclic AMP phosphodiesterases of *Dictyostelium discoideum*. *Biochim. Biophys. Acta* **758**, 49–57.
- Shaulsky, G., Kuspa, A., and Loomis, W. F. (1995). A multidrug resistance transporter serine protease gene is required for prestalk specialization in *Dictyostelium*. *Genes Dev.* **9**, 1111–1122.
- Shaulsky, G., Escalante, R., and Loomis, W. F. (1996). Developmental signal transduction pathways uncovered by genetic suppressors. *Proc. Natl. Acad. Sci. USA* **93**, 15260–15265.
- Siebert, F., and Weijer, C. (1989). Digital image processing of optical density wave propagation in *Dictyostelium discoideum* and analysis of the effects of caffeine and ammonia. *J. Cell Sci.* **93**, 325–335.
- Siebert, F., and Weijer, C. J. (1992). Three-dimensional scroll waves organize *Dictyostelium* slugs. *Proc. Natl. Acad. Sci. USA* **89**, 6433–6437.
- Siebert, F., and Weijer, C. J. (1995). Spiral and concentric waves organize multicellular *Dictyostelium* mounds. *Curr. Biol.* **5**, 937–943.
- Siebert, F., Weijer, C. J., Nomura, A., and Miike, H. (1994). A gradient method for the quantitative analysis of cell movement and tissue flow and its application to the analysis of multicellular *Dictyostelium* development. *J. Cell Sci.* **107**, 97–104.
- Steinbock, O., Siebert, F., Muller, S. C., and Weijer, C. J. (1993). Three-dimensional waves of excitation during *Dictyostelium* morphogenesis. *Proc. Natl. Acad. Sci. USA* **90**, 7332–7335.
- Traynor, D., Kessin, R. H., and Williams, J. G. (1992). Chemotactic sorting to cAMP in the multicellular stages of *Dictyostelium* development. *Proc. Natl. Acad. Sci. USA* **89**, 8303–8307.
- Wang, B., and Kuspa, A. (1997). *Dictyostelium* development in the absence of cAMP. *Science* **277**, 251–254.
- Williams, J., and Morrison, A. (1994). Prestalk cell-differentiation and movement during the morphogenesis of *Dictyostelium discoideum*. *Prog. Nucleic Acid Res. Mol. Biol.* **47**, 1–27.
- Wu, L., Franke, J., Blanton, R. L., Podgorski, G. J., and Kessin, R. H. (1995). The phosphodiesterase secreted by prestalk cells is necessary for *Dictyostelium* morphogenesis. *Dev. Biol.* **167**, 1–8.
- Xiao, Z., and Devreotes, P. N. (1997). Identification of detergent-resistant plasma membrane microdomains in *Dictyostelium*: Enrichment of signal transduction proteins. *Mol. Biol. Cell* **8**, 855–869.

Received for publication June 16, 1997

Accepted August 19, 1997